Research: Workflow Management Systems

A workflow describes the work needed to produce an outcome, who or what performs it, and the rules controlling its progress. A workflow management system stores these descriptions and manages their execution. A workflow definition describes the process, while a workflow instance is one execution of it. Each instance needs its own data, participants and progress. TIVIDY should therefore separate the reusable definition from information that changes during execution. Definitions typically contain activity descriptions, routing rules, required roles, and validation constraints, while instances contain the actual data values, assignments, progress, and history for that execution [1], [2].

Workflows involve more than a list of tasks. Activities may happen in sequence, run in parallel, or become available only when a condition is satisfied. Research identifies these as recurring workflow behaviours. BPMN, a standard for describing business processes, also represents individual tasks and smaller processes within larger ones [3]. These findings suggest that TIVIDY needs to represent grouped work and track the conditions that make each activity ready.

Routing determines which activity happens next. Assignment determines who performs it. Workflow systems typically model participants through roles or groups, with assignment rules determining which eligible participant receives a work item [2]. Camunda, an existing workflow-management platform, supports tasks assigned to a specific user or made available to eligible users or groups [4]. Approval, rejection and escalation also need defined rules. Escalation means passing work or a problem to another responsible person or level. TIVIDY should allow these policies to change without rewriting its basic task classes. Routing, assignment, approval and escalation rules are typically configurable rather than hard-coded, allowing organisations to change policies without modifying the workflow engine [1].

Workflow progress can also change when something happens, such as a task completing, a message arriving or a deadline passing. BPMN represents these occurrences as events [3]. Camunda's job workers perform automated tasks and report success or failure. Failed work can be retried, with a delay between attempts, before requiring intervention [5]. TIVIDY therefore needs clear responsibilities for events, external-system communication and failure handling. Integration with external systems is often handled through defined interfaces or connectors, so that changes to external services do not affect the core engine [5].

A workflow system must explain both its current progress and its past actions. An audit history records what happened during an instance. Camunda's Operate tool illustrates how users can inspect active and completed instances and investigate problems [7]. We propose that TIVIDY record actions and state changes, then use that information to show progress, overdue tasks and failures.

Changes to process definitions also need control. Camunda normally keeps existing instances on the definition version they started with [6]. TIVIDY should preserve that

association. Recovery also has limits: cancelling unfinished work differs from reversing completed work. BPMN uses compensation for actions intended to reverse completed effects [3]. Restoring internal data alone cannot reverse every external action.

Together, these findings suggest a generic engine with separate responsibilities for definitions, instances, work items, participants, decisions, events and history. The selected scenarios should supply their own activities, data and policies. Routing rules, assignment policies, approval and escalation logic, and external integrations are expected to vary independently from the core engine. These responsibilities must collaborate before the team makes final design pattern choices and designs the UML diagrams.

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